

**Govt. M.A.O. Graduate College, Lahore.**

MIDTERM EXAMS (May 2025)

Department/Discipline: Physics

Paper: Calculus II

Course code: MPHY-102

Teacher's Name: Syed Asif Iqbal Kazmi

Subject: Calculus

SEMESTER: II (Morning)

Max. Marks: 25

Total Time: 1:00 Hour

Student's Name: ABOUL-RAUF College Roll No. 24120 PU Roll No. 127884**NOTE:** Attempt all questions**Q1:** Find Arc length of curve (in cylindrical coordinates)

$$r = e^{2t}; \theta = t; z = e^{2t}; \quad 0 \leq t \leq \ln 2 \quad [5]$$

Q2: show that the lines $x = 1 + 7t; y = 3 + t; z = 5 - 3t$ and $x = 4 - t; y = 6;$ $z = 7 + 2t$ are skew lines. Also find the shortest distance between them. [10]**Q3:** Locate the point of intersection of the plane $2x + y - z = 0$ and the line through point $(3, 1, 0)$ that is perpendicular to the plane. [5]**Q4:** (a) Convert from rectangular coordinates $(1, -1, \sqrt{2})$ to spherical coordinates.(b) Identify the surface $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ Also find its intercepts. [2.5+2.5]